

Year 2 Assessed Problems

Semester 1

Assessed Problems 3

SOLUTIONS TO BE SUBMITTED

ON CANVAS BY

Wednesday 29th October

at 17:00

Year 2 Particles and Nuclei
Problem Set 1

Problem 1 The J/ψ is a meson consisting of a bound charm-quark anti-charm-quark pair, discovered in 1974 at both SLAC and Brookhaven National Laboratory in the USA. The radius of the J/ψ is ≈ 0.2 fm. The spin of the J/ψ is 1, hence the J/ψ is a boson, albeit not a fundamental boson.

Estimate the minimum velocity of the charm-quark in the J/ψ meson. Is the charm-quark velocity within the J/ψ relativistic?

What would be the range of a force produced by the exchange of a J/ψ meson?

[5 points]

Year 2 Particles and Nuclei
Problem Set 2

Problem 1 SLAC discovered the structure of the proton by scattering an electron beam off a proton at a centre-of-mass energy of 6.4 GeV.
If the proton is at rest, what is the momentum of the electron beam, assuming the electron momentum is in the positive z -direction?
[5 points]